

If the primal is a minimization problem, and the dual is written as a maximization problem, then the following rules apply.

- For each constraint of the primal there is a dual variable. The sign of the dual variable depends on the sign of the primal constraint, according to the following table:

Constraint of the primal	Dual variable
$=$	free
$\leq$	≤ 0
$\geq$	≥ 0

- For each primal variable there is a dual constraint. The sign of the constraint depends on the sign of the primal variable, according to the table below.

Primal variable	Dual constraint
≥ 0	$\leq$
≤ 0	$\geq$
free	$=$

Write the dual of the following linear optimization problem in two ways: first use the summary tables above, and then use the Lagrangian method in order to see the correctness of those summary tables.

$$\min -4x_1 - 7x_2$$

subject to

$$-5x_1 + 4x_2 \leq 16$$

$$x_1 + 3x_2 \leq 31$$

$$x_1 + 2x_2 \leq 24$$

$$3x_1 + 5x_2 \leq 68$$

$$x_1 + x_2 \leq 122$$

$$x_1 \geq 0$$

$$x_2 \geq 0.$$